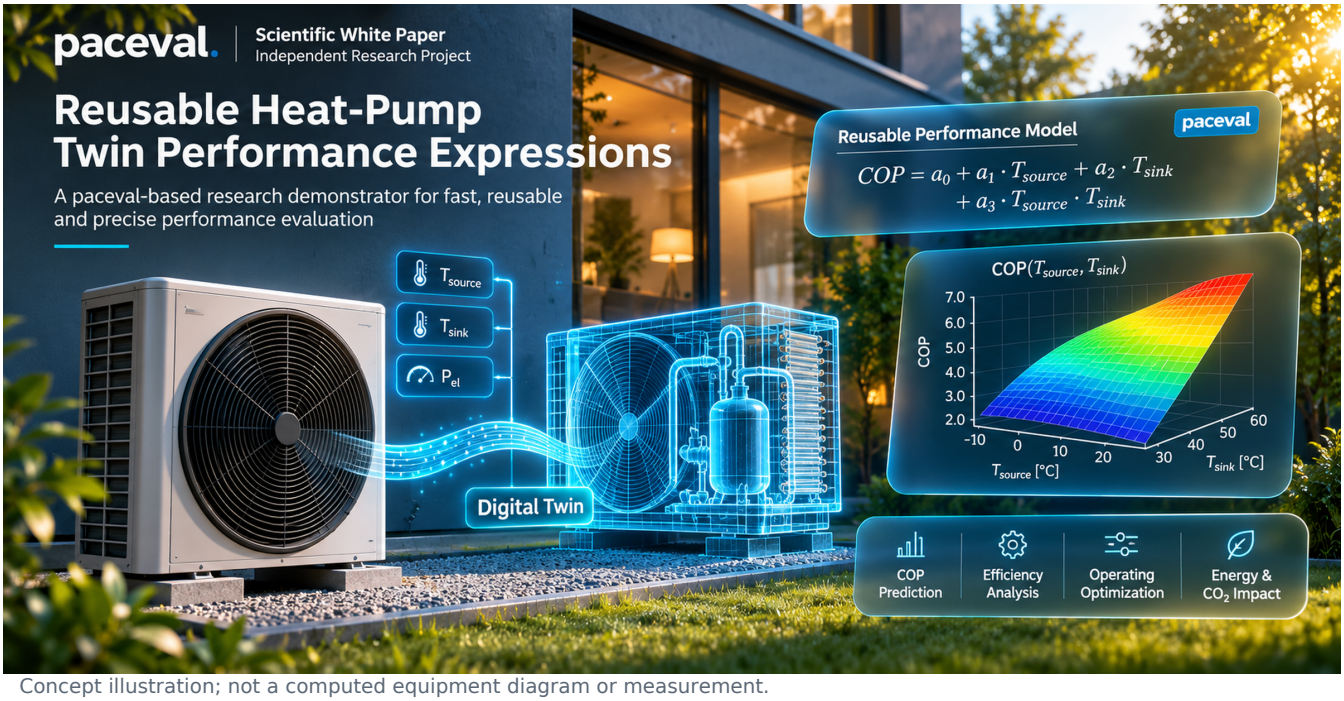
Reusable Heat-Pump Twin Performance Expressions

A reproducible, batch-driven C++/paceval study of synthetic heating performance maps, composite system metrics and formula versioning



Concept illustration; not a computed equipment diagram or measurement.

Executed research outcome. Twenty-three persistent mathematical graph definitions and 6,000 synthetic operating candidates were evaluated using the original paceval Linux engine. Their composite thermal, electrical, economic and constraint expressions were independently audited.

Scientific boundary. The results show expression-execution feasibility, not physical calibration of a specific heat pump, validated frost behavior, measured seasonal efficiency or certification to EN 14511 / EN 14825.

1. Introduction: industrial use case and state of the art

Buildings increasingly rely on electric heat pumps for low-carbon heating. Equipment developers and building-energy engineers must estimate how heating capacity and electricity consumption vary with the outdoor source temperature, the sink-water delivery temperature, partial loading and building heat demand. A digital twin may use these performance maps repeatedly for planning, supervisory analysis and operational what-if studies.

A heat pump transfers heat from a source to a higher-temperature sink. Its coefficient of performance (COP) is the ratio of useful heating to input electrical power at a defined operating point. Unlike an idealized thermodynamic cycle, real equipment also depends on compressor speed, heat exchangers, auxiliaries, frost and cycling, and cannot be certified by a synthetic algebraic map.

Established and recent methods

EnergyPlus documents equipment performance relationships based on temperature-dependent capacity and energy-input-ratio curves. Modelica and FMI support physical modeling and co-simulation. A 2026 Applied Energy systematic review found that digital-twin model choice must account for robustness, generalizability and integration, while a 2026 IEA Heat Pumping Technologies experimental comparison highlighted how dynamic cycling and defrost behavior can differ from selected standard test points.

Research question

Can the full set of repeatable algebraic performance-map calculations, not only a small COP equation, be registered as reusable and versioned paceval graphs, evaluated through authentic scalar and batch API calls, and independently verified in an industrially interpretable workflow?

Experiment input and output

Operating inputs	Model outputs
Source and sink temperatures, load ratio, nominal capacity	COP, available and delivered heat, electricity
Thermal demand, auxiliary power, tariff, time	Heat margins, electrical bounds, costs, sensitivities

All coefficients and operating profiles in this report are synthetic. No actual manufacturer or client equipment supplied calibration or monitoring data.

2. Research architecture and mathematical ownership

The host C++ application generates validated operating points, manages the real paceval library, retains calculation handles, executes official Multi-Graph-Batch calls and writes deterministic result tables. It also evaluates explicit acceptance rules and ranks the computed candidate scores. No external numerical engine generates the reported scientific outputs.

Processing stage	Actual executor
Input parsing, permitted ranges, deterministic scenarios	C++ host
COP V1/V2, heating capacity, delivered heat	paceval
Part-load and synthetic cold-weather factors	paceval
Power, system COP, source heat, energy, electricity cost	paceval
Three signed operating margins and two policy scores	paceval
Two analytical raw-COP sensitivities	paceval
Select lowest score, write CSV, aggregate replay energy	C++ host
Independent reference mathematics	Native C++ / mpmath only for verification

Reusable computation graphs

Twenty-two operating-map functions and one aggregate replay energy-ratio function are each created once and preserved as live SDK handles while a large sequence of inputs is evaluated. The graph registry and a function-level execution inventory are included, establishing an auditable distinction between paceval mathematics and host orchestration.

Why this division matters

The host does not recompute significant domain results after receiving arbitrary intermediary graph values: the full delivered-heat, electric-power, system-COP, constraint and cost expressions are submitted directly to paceval. Stateful candidate ranking and temporal summation remain host tasks.

3. Defined synthetic mathematical operating model

Temperatures in degrees Celsius; all heat and electrical rates in kW. The coefficients define a hypothetical heating-mode example within -15 to +20 °C source, +30 to +60 °C sink and 0.25 to 1.0 load ratio. No claimed parameter set belongs to an actual heat pump.

3.1 Temperature-dependent reference COP

$$C_1 = 4.4 + 0.065T_s - 0.045(T_k - 35) - 0.0006(T_k - 35)^2 + 0.002T_s(T_k - 35)$$

Version 2 changes the base, source-temperature and interaction coefficients. Both versions remain registered simultaneously and are sampled at the same inputs.

3.2 Capacity and part-load models

$$F_{cap} = 1 + 0.014T_s - 0.005(T_k - 35)$$

$$Q_{avail} = Q_{rated}F_{cap}, \quad Q_{heat} = \lambda Q_{avail}$$

$$F_{PLR} = 0.82 + 0.18\lambda - 0.06(1 - \lambda)^2$$

3.3 Declared hypothetical cold-weather modifier

$$F_{cold} = 1 - 0.08\exp\left(-\left(\frac{T_s - 1}{3}\right)^2\right)$$

The cold-weather modifier is deliberately synthetic. Its shape is useful for demonstrating nonlinear expression reuse but is not a measured defrost penalty. The model contains no dynamic frost, refrigerant, heat-exchanger or control state.

4. Heat balance, power, margins and sensitivities

The model returns complete system-side algebraic results using paceval, not merely the COP polynomial.

$$C_{eff} = C_1 F_{PLR} F_{cold}$$

$$P_{comp} = Q_{heat}/C_{eff}, \quad P_{el} = P_{comp} + P_{aux}$$

$$C_{system} = Q_{heat}/P_{el}, \quad Q_{source} = Q_{heat} - P_{comp}$$

$$E_{heat} = Q_{heat}\Delta t, \quad Cost = P_{el}\Delta t\, c_{el}$$

4.1 Example screening margins

The three mathematically computed signed guards are heating margin = delivered heat minus requested load, electrical margin = 4.8 kW minus electric input, and COP margin = system COP minus 2.5. The host interprets nonnegative margins as satisfying the illustrative screening rule; this is not physical equipment protection logic.

4.2 Explicit derivatives of the raw polynomial

$$\frac{\partial C_1}{\partial T_s} = 0.065 + 0.002(T_k - 35)$$

$$\frac{\partial C_1}{\partial T_k} = -0.045 - 0.0012(T_k - 35) + 0.002T_s$$

These derivatives are sensitivities of the stated COP map only. They are not derivatives of a time-dependent digital-twin system. Weighted composite policy scores V1/V2 are also registered mathematical expressions, with the independent host choosing the minimum.

5. Deterministic research workload and method

Workload dimension	Actual executed coverage
Synthetic operating candidates	6,000 validated cases
Distinct retained graphs	23 mathematical functions
Multi-graph candidate outputs	132,000 values
Temperature performance surface	3,721 (61 x 61) points
Seven-day operating replay	168 hourly points
Simultaneous formula version sweep	101 matched points
Host invalid-input regression	5 / 5 correctly rejected

Operating data and declared domains

Deterministic pseudorandom generation produces source temperatures between -13 and +18 °C and sink temperatures between +31 and +57 °C. Part-load, declared rated capacity, thermal demand, tariff, interval length and auxiliary input vary within the documented input contract. Surface and time-series data are generated by separate fixed sweeps, and every outcome is written to CSV at 17-significant-digit precision.

Batch correctness and lifecycle

The application checks graph construction and the result layout of the Multi-Graph-Batch API using a one-point scalar probe before executing independent batches. Handles remain alive for the full experiment and are deleted on shutdown; error flags and nonfinite results are checked. CTest runs a separate smaller workload with the original shared library, without any simulated replacement.

Independent checks

The released reference scripts separately calculate selected results using native C++ and 80-decimal-digit mpmath arithmetic, check the full composite output chain and validate domain and physical accounting identities. Detailed technical numerical differences reside in audit files, not in this published report.

6. Executed results and scientific status

Verified result	Recorded outcome
paceval retained mathematical graphs	23
Independent native scalar results	4,963 checked
Independent 80-decimal-digit calculations	880 checked
Synthetic candidates inside three example margins	2,613 of 6,000
Seven-day delivered heat	1,080.19 kWh
Seven-day electrical consumption	331.20 kWh
Seven-day scenario energy ratio	3.2615
Real-SDK CMake/CTest	1 / 1 passed

The energy ratio is calculated from the aggregated synthetic replay values. It is not a measured seasonal performance factor, a manufacturer COP claim or an EN 14825 SCOP result. It must not be applied outside the stated artificial scenario.

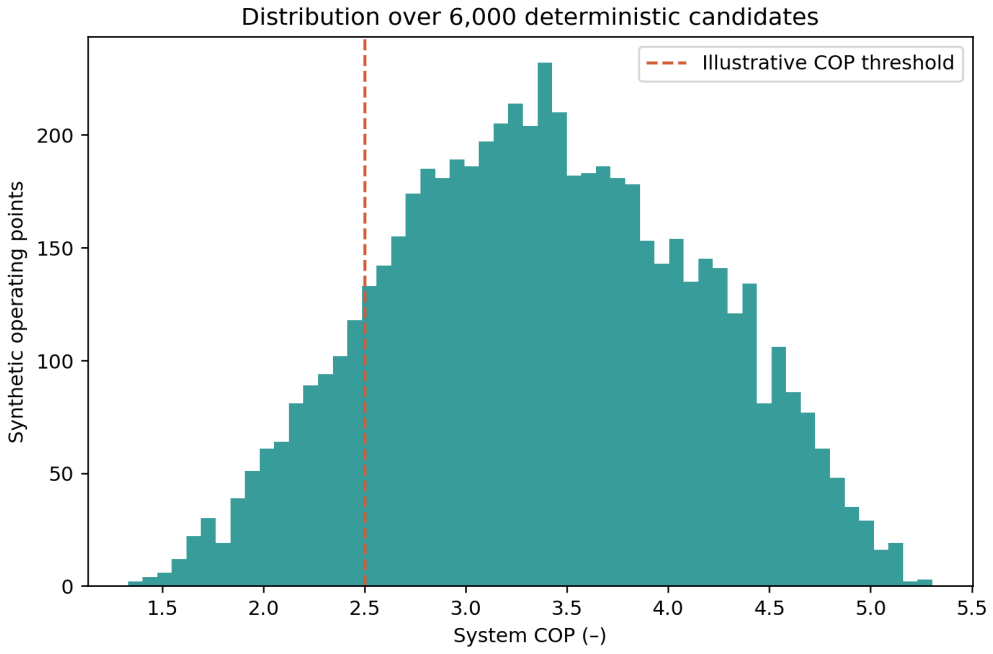


Figure 5. Distribution of system COP calculated at 6,000 deterministic synthetic operating points; the line marks an illustrative numerical threshold.

7. Industrial operating maps: COP and capacity

The temperature performance maps demonstrate a reusable algebraic representation that an external building simulation or equipment application could sample at alternative operating conditions. The surfaces are generated from paceval batch values; they are not laboratory performance maps.

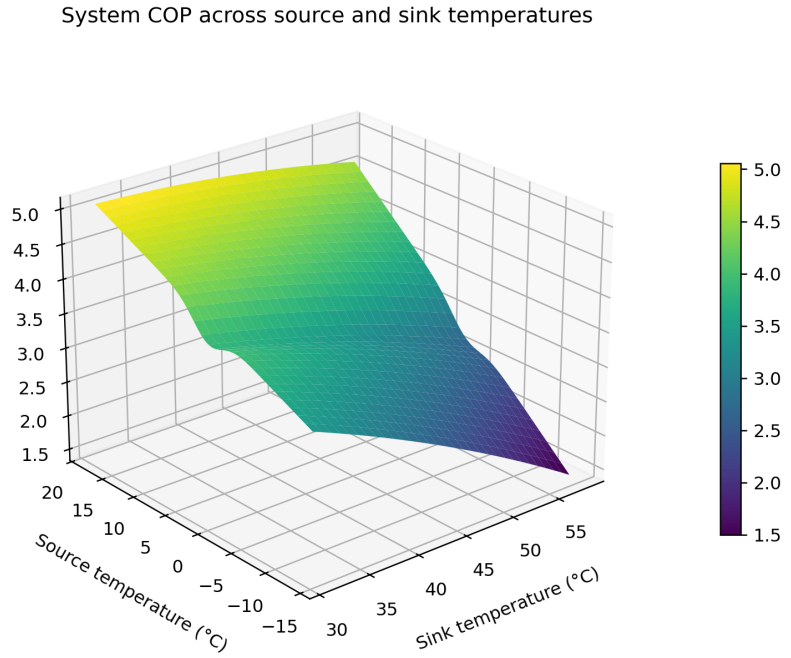


Figure 1. Three-dimensional system-COP surface, source and sink temperature inputs.

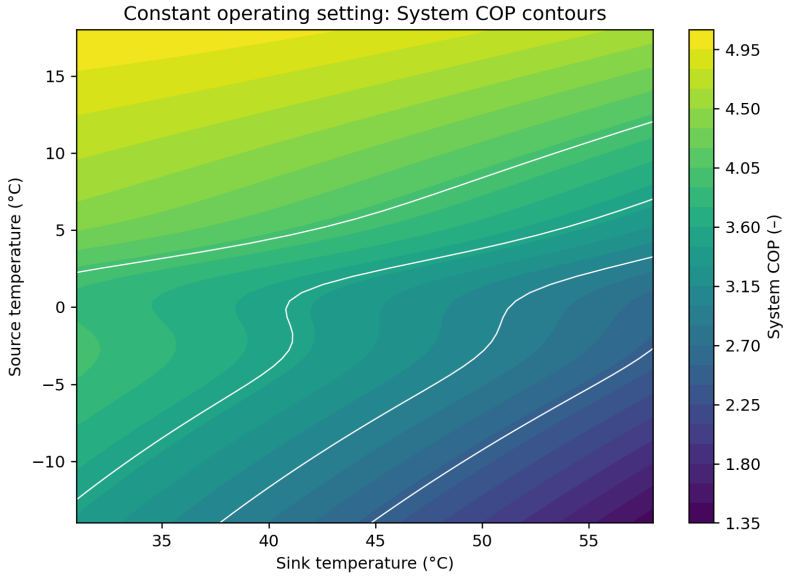


Figure 2. Matching two-dimensional COP contours, fixed part-load and auxiliaries.

At a fixed part-load setting, the hypothetical map generally favors warmer sources and lower sink-water temperature. The graph reveals the model specification rather than a universal behavioral law: real performance depends on actual equipment and its control strategy.

7. Industrial operating maps: capacity and electricity

Separating available heating capacity from electrical input helps a screening tool assess whether a model can satisfy load and power constraints at a declared operating point.

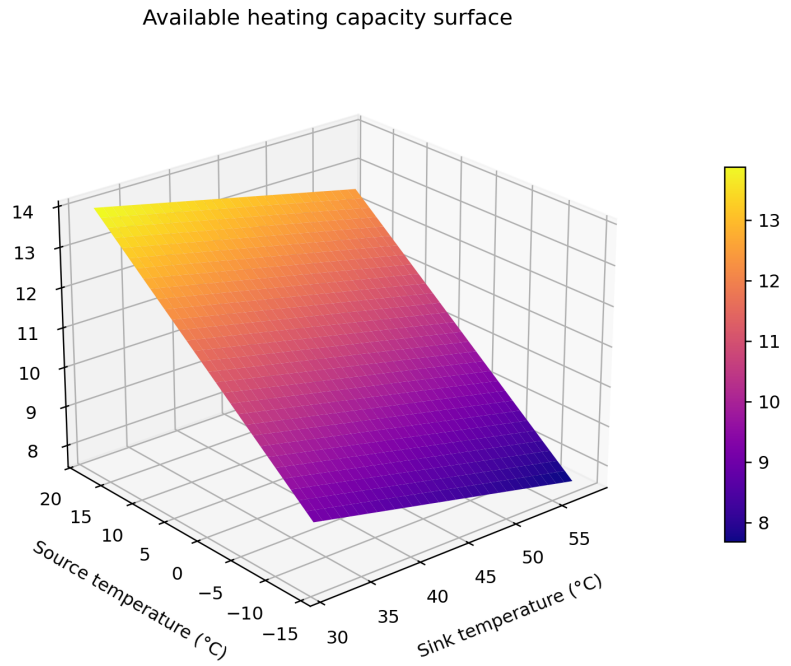


Figure 3. Declared capacity modifier and synthetic rated heating capacity across temperatures.

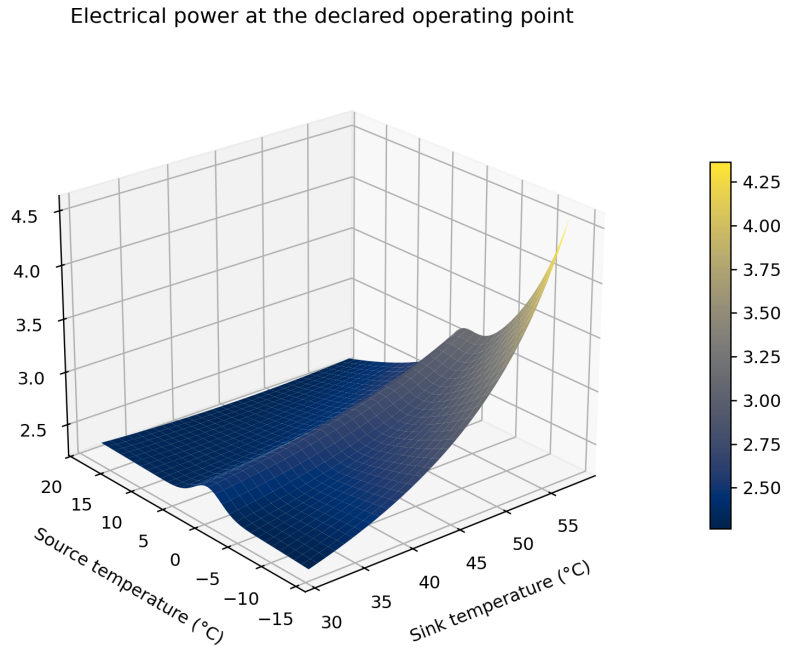


Figure 4. Electrical input, including the specified auxiliary power, derived by paceval.

Both quantities come from their own complete reusable paceval expressions. The C++ host does not substitute an unrelated performance calculation during visualization. No compressor map or device test specification is inferred from the plotted numbers.

7. Industrial scenarios: temporal replay and screening

The seven-day profile is a deterministic operating-point sequence, not a time-integrated physical heat-pump state. Each hour is evaluated independently from its input conditions, and the host aggregates the returned paceval energy and power results.

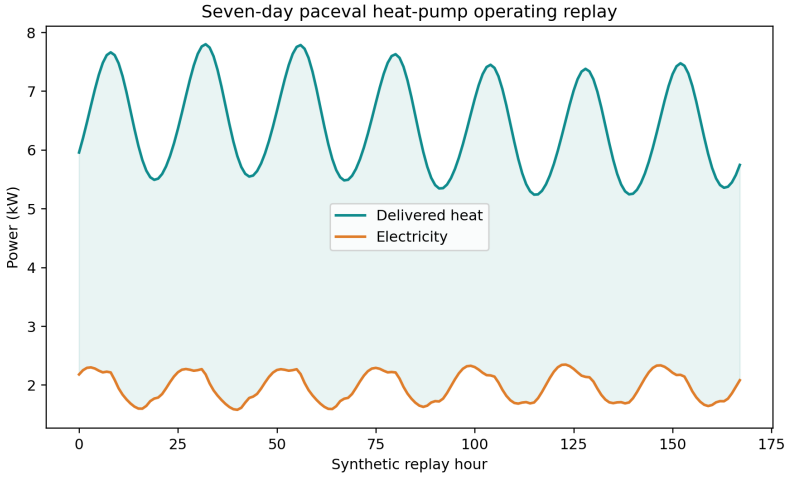


Figure 6. Hourly thermal delivery and electricity demand over a synthetic 168-hour replay.

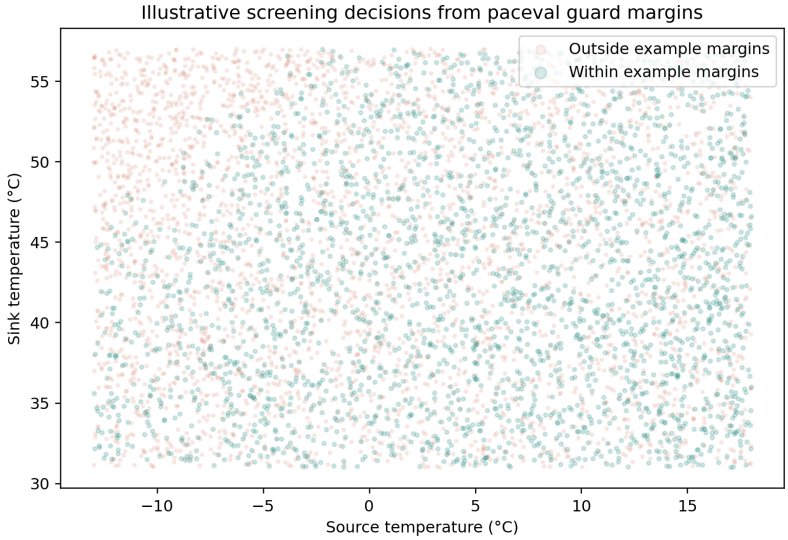


Figure 9. Which sampled input cases satisfy the three illustrative paceval guard margins.

A real supervisory system would also evaluate start/stop losses, transient building loads, refrigerant limits and safety interlocks. The current study verifies the mathematical policy and replay arithmetic, not operational safety.

8. Formula versions and analytical sensitivities

Two COP map variants and two weighted policy scores are preserved in independently accessible paceval graph handles. This demonstrates controlled mathematical model changes, while the host maintains version identifiers and candidate selection logic.

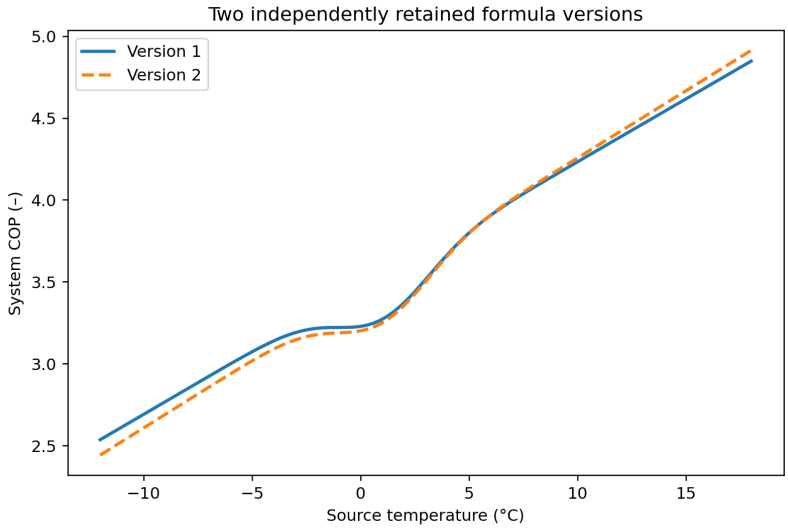


Figure 7. Both system-COP model versions evaluated at the same input sweep.

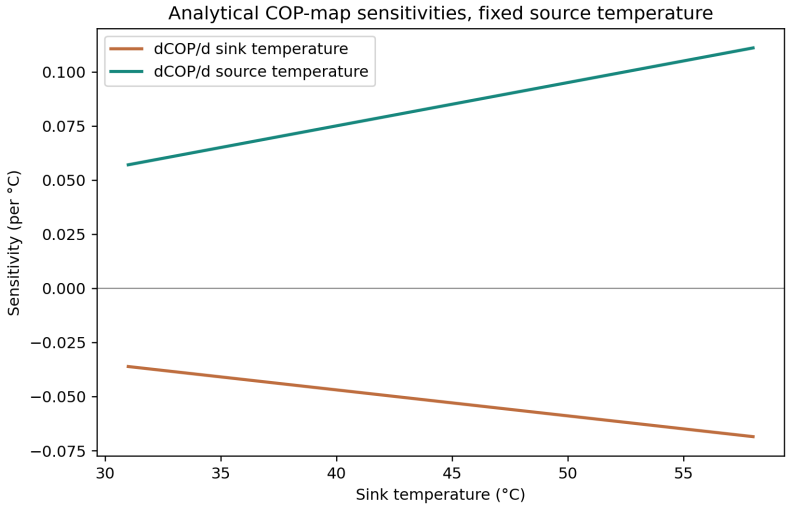


Figure 8. Two explicit derivatives of the raw COP polynomial, as paceval graphs.

These formulas offer a transparent basis for model-update regression tests and engineering what-if studies. They do not constitute automatic differentiation of an arbitrary algorithm, nor prove an industrial improvement over competing simulation engines.

9. Validation, source code and reproducibility

The original SDK interface files are distributed in src/ with their upstream notices intact. The runtime shared library is intentionally external. CMake uses the original headers and implementation, and the C++ program executes paceval graph creation, scalar checking and Multi-Graph-Batch evaluation before exporting five CSV result artifacts.

Release inventory

File or location	Reproducibility role
src/heat_pump_twin.cpp	Original SDK C++ workflow, formula graph registry and input contract
src/CMakeLists.txt	External shared-library build with actual integration CTest
examples/*.csv	Inputs, complete paceval outputs and formula-version sweeps
results/math_execution_inventory.csv	Function-level actual math-execution responsibility
src/high_precision_reference.py	Separate independent 80-digit numerical audit
src/verify_results.py	Composite energy and operating-margin consistency
src/make_figures.py / src/make_whitepaper.py	Regenerate plotted scientific figures and final PDF
src/check_readme_math.py	Static GitHub math and relative-link regression checks

Reproduction commands

Configure with cmake -S src -B build -DCMAKE_BUILD_TYPE=Release -DPACEVAL_RUNTIME=<absolute-library-path>; compile with cmake --build build -j, execute ctest --test-dir build --output-on-failure, then run ./build/heat_pump_twin --out . --cases 6000. Execute the included Python verification and plotting scripts as shown in the README.

10. Industrial interpretation and scientific limitations

What has been demonstrated

The research demonstrates the reuse of persistent mathematical expressions for physical-like operating maps and compound performance quantities. It confirms successful scalar and Multi-Graph-Batch execution, independent verification of the released results, explicit unit/domain contracts, temperature sensitivity formulas and simultaneous versioned model handling.

What has not been demonstrated

The synthetic coefficients are not fitted to real equipment. No measured refrigerant properties, compressor map, system storage, transient defrost physics, building control feedback, equipment degradation, electrical-grid interaction or independent certification experiment is included. A seven-day energy quotient is not a standards-based SCOP. The declared guards are research policies, not validated hardware or regulatory limits.

From kernel to industrial digital twin

A future application would need permitted test-bench and field datasets, documented model identification and uncertainty, measured frost/cycling behavior, a transient building/system solver, and model exchange or integration through an FMI/Modelica or EnergyPlus interface. A validation protocol should separate training/calibration points from unseen operation and quantify the errors of the entire operating process.

Research contribution

The contribution is intentionally modest and verifiable: comprehensive reuse of an algebraic calculation layer for a transparent engineering heat-pump scenario. No computational-speed advantage, reduced emissions, energy savings or guaranteed physical prediction accuracy is asserted.

11. Conclusion and external references

Conclusion. The defined heat-pump mathematical performance kernel was implemented with the original paceval engine and executed for 6,000 synthetic operating points. The study confirmed 23 persistent formulas, multi-graph batch reuse, composite thermal/electrical and cost evaluation, independent verification and version-controlled modifications. Its scientific interpretation is limited to the specified mathematical model, not the prediction of a particular manufacturer device.

Selected technical and scientific sources

1. Applied Energy, 2026: Modeling methods for chiller and heat pump systems in buildings: systematic review in the context of digital twins. doi:10.1016/j.apenergy.2026.127632

2. Renewable and Sustainable Energy Reviews, 2026: State-of-the-art of digital twin HVAC+R systems for buildings. doi:10.1016/j.rser.2025.116601

3. EnergyPlus Engineering Reference: Plant Loop Heat Pumps: performance temperature curves, capacity and electric-power calculations. energyplus.readthedocs.io

4. IEA Heat Pumping Technologies, 2026: Experimental comparison of seasonal efficiency using EN 14825 and dynamic load emulation, May 2026.

5. Functional Mock-up Interface: Specification for simulation and model exchange. fmi-standard.org/docs/main/

6. Modelica Association: Modelica Language Specification. specification.modelica.org

7. paceval SDK: Official open-source repository and C/C++ interface. github.com/paceval/paceval

Release boundaries

All plotted values are generated from synthetic model inputs and authentic paceval output files. The broad cover visualization is conceptual. The external shared library is not included. Independent checks and exact formula definitions are supplied so results can be independently reproduced and extended.

paceval | Reusable Heat-Pump Twin Performance Expressions | Research package WP_038 | 13 November 2026